

FLUTTER APPLICATION DEVELOPMENT

Semester Project — Student Instructions

Spring 2026 | National Textile University | BSCS

PROJECT OVERVIEW

Project Type	Flutter Mobile Application (Academic Semester Project)
Submission Date	To be announced by course instructor
Group Size	Maximum 2 Students per Group
Project Selection	Students have selected their project ideas today
Platform	Android / iOS using Flutter Framework

1. Group Formation Rules

All students must strictly follow the group formation policy stated below:

- **Each group must consist of a maximum of 2 students.**
- Solo (individual) submissions are also allowed.
- No group of 3 or more students will be accepted under any circumstance.
- Each student can only be part of ONE group.
- Once groups are finalized and submitted, changes will NOT be permitted.

⚠ Note: Projects selected today are considered FINAL. Students are responsible for ensuring their idea is unique and not a duplicate of another group's project.

2. Project Idea Requirements

Your selected project must fulfill the following criteria to be approved:

1. The application must be developed using the Flutter framework (Dart language).
2. The project must solve a real-world problem or fulfill a practical use case.
3. The project idea must be original — copied or previously submitted apps are strictly not allowed.
4. The app must have a proper UI with at least 5 distinct screens/pages.
5. The project must include backend integration (Firebase, REST API, or local database).

- All 6 core modules listed in Section 3 must be implemented.

3. Required Application Modules

Every submitted Flutter project must include the following modules. The first 6 are mandatory for passing.

#	Module Name	Key Responsibilities	Status
1	Authentication Module	User registration, login, logout, form validation, session management	✓ Required
2	User Interface (UI) Module	Screen layout, navigation, responsive design, dialogs, loading indicators	✓ Required
3	Data Management (CRUD)	Create, read, update, delete records; confirm delete actions	✓ Required
4	API / Backend Integration	HTTP requests, JSON parsing, error handling, secure communication	✓ Required
5	State Management Module	Manage app state using GetX, Bloc, Riverpod, or Provider	✓ Required
6	Navigation Module	Named routes, argument passing, back navigation, stack management	✓ Required
7	Database Module	Data models, queries, relationships (Firebase / SQLite / MySQL)	Optional
8	Validation & Error Handling	Input validation, error messages, network failure handling, crash prevention	Optional

Legend: Green rows = Mandatory for passing. White rows = Strongly recommended for higher marks.

4. Project Submission Requirements

Your final submission must include all of the following deliverables:

- Full source code uploaded to GitHub (public or shared repository).
- Working APK file of the application.
- Project Report (PDF format) covering all implemented modules.
- Presentation slides (minimum 10 slides) for demonstration day.
- Short demo video (2–5 minutes) showing the app running.
- Group registration form with student names and IDs submitted to CR.

5. Evaluation Criteria

Projects will be evaluated based on the following weightage:

Evaluation Criteria	Marks	Weight
Module Implementation (all 6 core modules)	30	30%
UI / UX Design Quality	20	20%
Backend & Database Integration	15	15%
Code Quality & Clean Architecture	10	10%
Project Report & Documentation	10	10%
Live Demo & Presentation	15	15%
TOTAL	100	100%

6. Important Policies & Deadlines

- Project ideas submitted today are considered final — no changes after confirmation.
 - Plagiarism of any kind (code or report) will result in ZERO marks for both groups.
 - All group members are equally responsible for the entire project.
 - Instructor reserves the right to ask any group member questions during the demo.
 - Late submissions will result in a deduction of 10 marks per day.
 - Use of AI-generated code must be declared in the report; full copy-paste is not allowed.
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For any queries, contact your Class Representative (CR) or course instructor.